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1. Consider the following four points in 3D space:

1 / 1 point

 $A(1, 2, 3)$, $B(3, 4, 5)$, $C(1, 1, 1)$, $D(2, 1, 3)$

Which of the following subsets of vectors created from these points consists of linearly dependent vectors?

☐ {A, C, D}☐ {A, B, D}☒ {A, B, C}☐ {B, C, D}✔ **Correct**

A set of vectors is dependent if a vector in the set can be represented as a weighted sum of the other vectors in the set. In the vectors described in the question, the set {A, B, C} is a set of linearly dependent vectors as we can represent vector B as a sum of vector A and 2 times vector C, that is: $B = A + 2C$.

2. Given three arbitrary linearly independent vectors

1 / 1 point

 $A(1, 2, 3)$, $B(2, 1, 3)$, $C(1, 2, 1)$.What would be the rank of the matrix $[A \ B \ C]$?☐ 0☐ 1☒ 3☐ Can not be determined✔ **Correct**

In linear algebra, the rank of a matrix is the maximum number of linearly independent column vectors (or rows, as the row rank is equal to the column rank) in the matrix. Therefore, the rank of a matrix containing 3 linearly independent 3D vectors will be 3. This is because each of the three vectors in 3D space is linearly independent, meaning that they cannot be represented as linear combinations of the other two.